

## ORIGINAL ARTICLE

# Incidence and molecular characterization of ESBL-producing and colistin-resistant *Escherichia coli* isolates recovered from healthy food-producing animals in Pakistan

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## Abstract

**Objectives:** To investigate the occurrence and molecular features of ESBL-producing and colistin-resistant *Escherichia coli* isolates recovered from healthy food-producing animals in Pakistan.

**Methods:** A total of 153 *E. coli* isolates were recovered from 250 faecal samples collected from livestock and poultry. The antibiotic susceptibility, resistant determinants and mobile genetic elements were determined for all the isolates. The clonal relatedness was analysed by MLST. Plasmids harbouring, localization and trans-ferability of *mcr-1* gene were carried out by Southern hybridization, S1-PFGE and transconjugation.

**Results:** Out of 153 *E. coli* strains, 49.01% isolates were ESBLs producers, whereas 18.95% were resistant to colistin and 84.31% of the isolates. Multidrug resistance was found in 84% of the isolates. The ESBL-producing *E. coli* in buffaloes, cattle, sheep, goat and broilers faecal samples were 60%, 74%, 54%, 50% and 68%, respectively. Among the ESBLs genes, *bla*<sub>CTX-M</sub> was the most prevalent group detected in 98.66%, while only *mcr-1* of the colistin-resistant genes could be PCR amplified in 29 isolates. The common MGEs found were *ISECP1* (35.13%), *ISCR1* (33.78%), *ISAp11* (20.27%) and *Int11* (58.10%). The most predominant Inc. types found were IncFIB 46.66%, followed by IncFIA 30.66%, IncFIC 26.66%, IncFrepB 26.66%, IncHI2 26.66%, IncP 22.66% and IncX4 21.33%. The most frequent sequence type detected was ST58. Southern blot and S1-PFGE confirmed the plasmid harbouring of *mcr-1* gene.

**Conclusion:** The co-occurrence of *mcr-1* and ESBLs-encoding genes, along with MGEs in *E. coli* from healthy food animals in Pakistan, is a major concern.

**Significance and Impact of Study:** Antimicrobial resistance can be transferred from animals to humans by direct contact or via the food chain and environment. The prevalence and co-occurrence of ESBL and colistin resistance genes from food-producing animals is rare in Pakistan. To our knowledge, this is the first report to find ESBLs and *mcr-1*-harbouring *E. coli* from the faecal samples of the healthy food-producing animals in Pakistan. The presence of ARGs in association with MGEs, co-harbouring the virulence factors, as determined in the current study, is a severe

threat to livestock and the human community as it has horizontally and food web transferability.

#### KEYWORDS

ESBL, *Escherichia coli*, food-producing animals, *mcr-1*, multidrug resistance

## INTRODUCTION

The rapid rise and spread of MDR Enterobacteriaceae, especially *Escherichia coli*, is increasing and has become a serious threat to public health and livestock. The extensive use of antibiotics in animal production affects the spread of MDR *E. coli* and has significantly led to the selection and spreading of resistant determinants in recent decades (Naylor et al., 2019). Antibiotics are added to animal feed as a prophylactic measure and growth enhancers at dosages ranging from 2.5 to 125 mg/kg for an indefinite duration (weeks to months) based on the size and types of the animals antibiotics (Marshall & Levy, 2011). Unfortunately, antibiotics usage in food-producing animals is not well regulated; hence, no estimation of its annual usage is available (Ur Rahman & Mohsin, 2019). Nevertheless, prevailed practices indicate that a major share of antibiotics usage is aimed for disease prevention and as growth promoters in Pakistan. On top of this, the availability of antibiotics over the counter encourages its overuse and accumulating pool of MGEs in the environment (Ali et al., 2021; Mohsin et al., 2017, 2019; Wang et al., 2020) that contribute to the emergence and dissemination of MDRs.

In the last decades, extended-spectrum  $\beta$ -lactamases (ESBL)-producing *E. coli* have been isolated from community-acquired extra-intestinal infections in humans, animals and food animals product, including beef, pork and chicken meat globally (Ali et al., 2020; Coque et al., 2008; Peirano & Pitout, 2010; Shafiq et al., 2019; Shafiq, Huang, Shah, Ali, et al., 2021). ESBL-producing *E. coli* shows resistance to extended-spectrum cephalosporins by hydrolysing their  $\beta$ -lactam rings. The most common ESBLs encoding genes are *bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub> and *bla*<sub>TEM</sub>. Of these, the *bla*<sub>CTX-M</sub> has been further categorized into five sub-groups (*bla*<sub>CTX-M-1</sub>, *bla*<sub>CTX-M-2</sub>, *bla*<sub>CTX-M-8</sub>, *bla*<sub>CTX-M-9</sub> and *bla*<sub>CTX-M-25</sub>) and more than 160 variants have been documented (<http://www.lahey.org/studies>). Food-producing animals have been one of the major sources of CTX-M-lactamases dissemination in the community (Meunier et al., 2006; Weissman et al., 2013; Zheng et al., 2012).

Colistin is considered one of the last therapeutic options against MDR pathogens; however, due to its overuse in animal husbandry, many resistance mechanisms developed against them (Liu et al., 2016). Initially, only

the chromosomal resistance to colistin was known, which are not much threat because of their only vertical transmission (Zhai et al., 2020). However, in November 2015, Liu et al. (2016), for the first time, reported the plasmid-mediated colistin-resistant gene *mcr-1* from China from both animal and human samples sources. Until now, *mcr-1* gene has been identified in 47 different countries from different origins and sources (Nang et al., 2019). Besides this, other *mcr* genes (*mcr-2* to *mcr-10*) are reported in different Enterobacteriaceae members (Wang et al., 2020). The mobilized colistin-resistant genes were initially identified on IncI2 and IncI4 plasmids, but later on, it was identified in different plasmid incompatibility types. Moreover, the insertion sequences such as IS*Apl2* and IS*Ecp1* were identified in association with *mcr* genes, enhancing their intergeneric and interspecies mobilization (Nang et al., 2019).

Based on the hazard associated with the miss and overuse of colistin, many developed countries have banned its use in animals husbandry as a growth promoter and advised other partner countries to lessen the practice of colistin in livestock (EMA Committee for Medicinal Products for Veterinary Use (CVMP) & EFSA Panel on Biological Hazards (BIOHAZ), 2017; Walsh & Wu, 2016). However, in Pakistan, the antimicrobial usage in food-producing animals is inordinate, and up to yet, there is no estimation of their annual usage. Pakistan national surveillance antibiotic resistance data have vast gaps, particularly on the extent, quality, quantity and pattern of antibiotic usage. Some recent studies reported *mcr-1* in *E. coli* from poultry origin in Pakistan, presenting the hazard due the misuse of antibiotics. (Azam et al., 2019; Lv et al., 2018). The dissemination of ESBLs and *mcr-1* is mainly linked with MGEs, including insertion elements (IS*CR1*, IS*Ecp1* and IS*Apl1*), integrons, and predominantly with the incompatible plasmids, which are believed to be the primary key factors of horizontal gene transfer (HGT) (Shafiq, Huang, Shah, Ali, et al., 2021; Wu et al., 2018).

The co-occurrence of ESBL and *mcr* genes adds further complications as many conjugative plasmids can carry multiple resistance genes by a single event of HGT (Yoon et al., 2018). It is highly anticipated that food-producing animals play a significant role in propagating these resistant determinants in the community through HGT and the food web. However, comprehensive studies describing the prevalence and co-occurrence of ESBL and colistin resistance genes from food-producing animals are rare in

Pakistan. Thus, this area should be further investigated to constrain the spread of ESBLs and colistin resistance genes from different animal origins. Therefore, in the current study, we aimed to screen *E. coli* isolates recovered from healthy food animals for ESBL and colistin resistance genes *mcr* (1–10). Furthermore, the characterization of various MGEs was performed to identify the potential animal and public health threats.

## MATERIALS AND METHODS

### Sampling of livestock faeces

A total of 250 fresh faeces were collected from different healthy animals, including buffaloes, cattle, sheep, goats and broilers from five local farms (50 samples from each farm) in Peshawar city of Khyber Pakhtunkhwa province, Pakistan. Briefly, the original faecal samples were collected using sterile cotton swabs from the rectum of buffaloes, cattle, sheep, goats and broilers' fresh droppings with the help of a registered veterinarian or animal attendant. All the collected samples were refrigerated immediately at 4°C and transported to the laboratory and processed within 4 h. The types of animals and the approaches used in this study are presented in Figure 1.

This work was approved by the ethical committee of the College of Veterinary Sciences, and Animal Husbandry, Abdul Wali Khan University, Mardan-Pakistan and all work prescribed here was performed under the departmental and national guidelines of handling and using animals for data collection. Proper consent from the owner of the animals or farms was obtained prior to sampling, and the procedure was also discussed.

### Microbiology

All faecal swabs were diluted 1:1 with sterile phosphate-buffered saline enriched with tryptic soy broth containing 20 µg cefotaxime, mixed vigorously, and incubated for 24 h in a shaker (120 rpm) at 37°C. Following incubation, 100 µl aliquots were plated on MacConkey agar and incubated for 18–24 h at 37°C. Bacterial growth was observed, and a single colony of presumptive ESBL *E. coli* was picked from each plate for further confirmation. All the putative ESBL *E. coli* colonies were confirmed biochemically and microscopically (Versalovic et al., 2011). To further confirm phenotypically, all the putative ESBL-producing *E. coli* isolates were screened by a double-disc synergy test according to Clinical and Laboratory Standards Institute (CLSI) guidelines (CLSI, 2017). All isolates were cryopreserved in Luria-Bertani (LB) broth medium supplemented

with 30% glycerol as stocks and kept at –80°C for further use.

### Molecular confirmation of *E. coli*

The preserved isolates were streaked on MacConkey agar to confirm the purity of the stocks, and a single colony was picked and grown in LB broth overnight at 37°C. The total genomic DNA was isolated using a bacterial DNA extraction kit (QIAamp DNA Microbiome Kit). The extracted DNA was used as a template to confirm *E. coli* by PCR using the *uidA* gene-specific primers, as defined earlier (Shafiq et al., 2019).

### Antimicrobial susceptibility testing

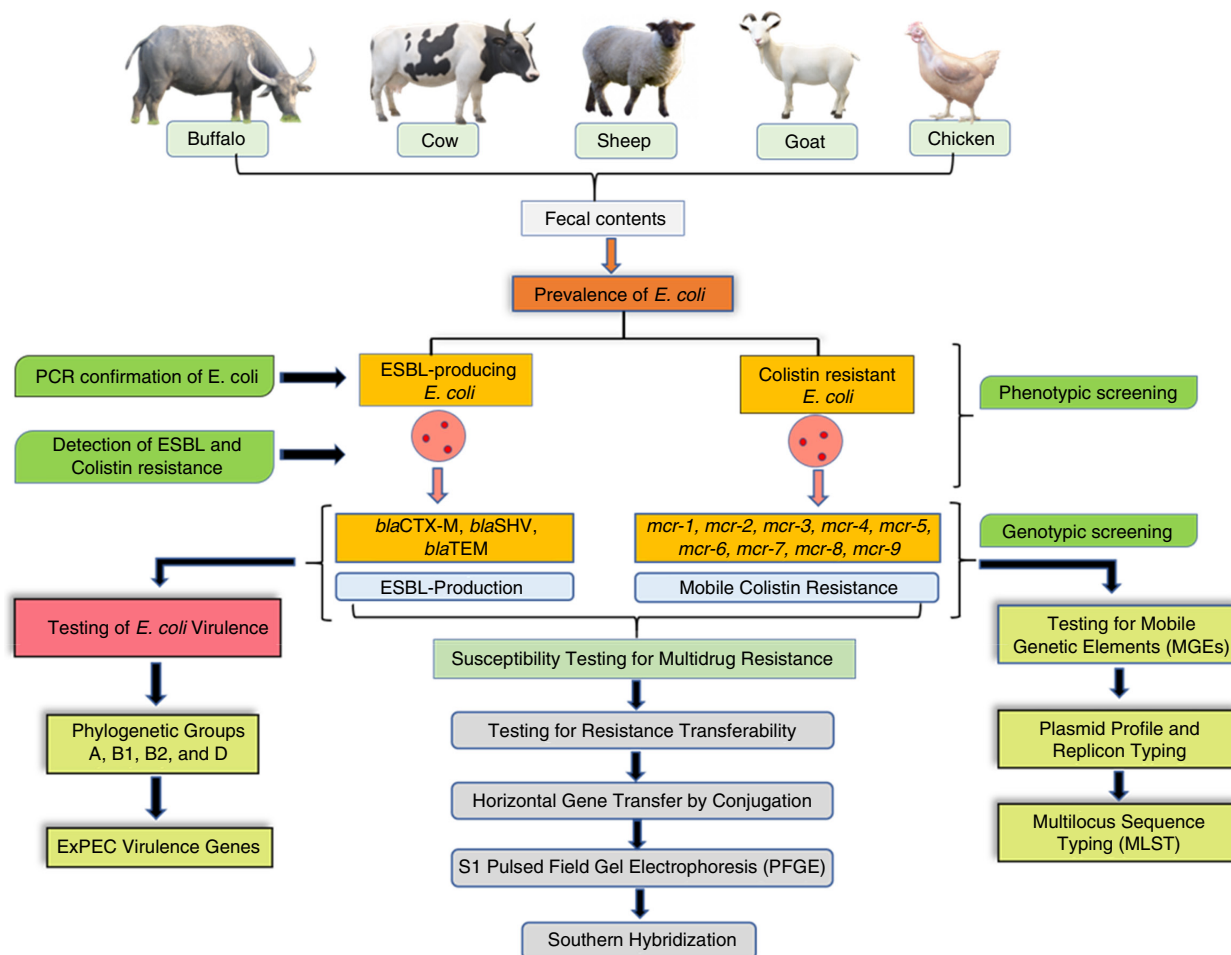
Antibiotic susceptibility was determined for each *E. coli* isolate using the broth microdilution technique to calculate the minimum inhibitory concentrations (MICs) to the corresponding antimicrobial agents according to the CLSI M100-ED28 guidelines (CLSI, 2017). Susceptibility testing involved the use of 13 different commonly used antibiotics, that is, ampicillin (AMP), cefoxitin (CFX), ceftiofur (CEF), cefotaxime (CTX), ciprofloxacin (CIP), chloramphenicol (CHL), colistin (COL), enrofloxacin (ENR), gentamicin (GEN), meropenem (MEM), polymyxin-B (Poly-B), tetracycline (TET) and trimethoprim (TRI). All the antimicrobials used were purchased from Sigma Aldrich Trading Co. Ltd. ATCC 25922 (colistin-sensitive *E. coli*) and ATCC 25933 (colistin-resistant *Proteus mirabilis*) were used for quality control.

### Extraction of genomic DNA

Each colony of *E. coli* was enriched overnight in 1.5 ml Luria-Bertani broth at 37°C, and DNA was isolated using a commercially available bacterial DNA extraction kit (QIAamp DNA Microbiome Kit). The quality and quantity of whole genomic DNA were measured using a Nanodrop OD-1000 spectrophotometer (Thermo-scientific®). The genomic DNA was further used to determine the presence of multiple resistant determinants and mobile-genetic elements (MGEs).

### Screening of ESBL and colistin resistance genes

Screening of ESBL-encoding genes (namely *bla*<sub>CTX-M</sub>, *bla*<sub>SHV</sub> and *bla*<sub>TEM</sub>) and colistin-resistant genes *mcr* (1–10)



**FIGURE 1** Summary of different approaches and animals used in this study

was performed by conventional PCR. All ESBL-positive *E. coli* isolates were further analysed for the presence of common *bla*<sub>CTX-M</sub> groups (namely *bla*<sub>CTX-M-1</sub>, *bla*<sub>CTX-M-2</sub>, *bla*<sub>CTX-M-9</sub> and *bla*<sub>CTX-M-25</sub>), as defined previously (Shafiq, Huang, Shah, Wang, et al., 2021). The details of targeted genes, PCR conditions, the primers and the amplicon sizes (bp) used in this study are summarized in Table S1. The amplified product was run on 1.2% agarose gel (Sigma Aldrich) placed in 1X Tris-acetate-EDTA (TAE) buffer by electrophoresis at 120 V for 25 min. Amplicons were visualized under the ultraviolet (UV) rays in the gel-Doc XR system (Bio-Rad) and subsequently sequenced to identify the target genes precisely.

### Screening of mobile genetic elements (MGEs) and plasmid replicons

To determine the presence of various MGEs involved in the transfer of ESBLs and colistin-resistant genes, different markers such as *ISEcp1*, *ISCR1*, *ISApII-mcr-1* and integrons (*Int1*, *Int2* and *Int3*) were checked through PCR using

target primers, as reported earlier (Shafiq, Huang, Shah, Ali, et al., 2021). To identify plasmid replicon types in ESBL- and *mcr-1*-positive *E. coli* isolates, PCR-based replicon typing (PBRT) KIT 2.0 (MBK0078; Diatheva, Cartoceto, Italy) was used for the characterization of 30 different plasmids. Eight separate multiplex PCRs were run, and the products were examined according to the kit instructions.

### Phylogenetic grouping and screening for virulence genes

According to Clermont et al. (2000), PCR-based phylogenetic grouping was determined by targeting five genes (*ChuA*, *yjaA*, *arpA*, *trpA* and an unspecified DNA component (TSPE4.C2) and grouped into four main phylogenetic classes (A, B1, B2 and D). All of these strains were then tested for the presence of different diarrheagenic *E. coli* (DEC) virulence genes, including enteroaggregative *E. coli* (EAEC), enterohemorrhagic *E. coli* (EHEC), enteroinvasive *E. coli* (EIEC), enteropathogenic *E. coli* (EPEC), diffusely adherent *E. coli* (DAEC) and enterotoxigenic *E.*

*coli* (ETEC) as described earlier (Shafiq, Huang, Shah, Ali, et al., 2021).

## Multi-locus sequence typing (MLST)

Multi-locus sequence typing (MLST) was performed to evaluate the genetic affiliation of the studied *E. coli* isolates, with seven housekeeping genes (*adhA*, *fumC*, *gyrB*, *icd*, *mdh*, *purA* and *recA*) were amplified by PCR according to the protocol previously defined (Wirth et al., 2006). After amplification, all the PCR products were sequenced following the guidelines of the MLST databases (<http://mlst.warwick.ac.uk/mlst/dbs/Ecoli/>, accessed on 9th September 2020). The sequence outcomes were used to obtain allelic numbers using an online database ([https://pubmlst.org/bigdb?pubmlst\\_escherichia\\_seqdef](https://pubmlst.org/bigdb?pubmlst_escherichia_seqdef), accessed on 9th September 2020).

All the allelic profiles and corresponding sequence types (STs) were assigned with BioNumerics software (Version 8.0; Applied Maths). A minimum spanning tree was built to visualize the genomic relatedness of strains isolated from different strains of food animals.

## Transconjugation

Conjugation experiments were carried out to determine the transferability of *mcr-1*-positive *E. coli* isolates. Six isolates bearing the *bla*<sub>CTX-M</sub> gene and the *mcr-1* gene were analysed by performing a transconjugation experiment. Azide-resistant *E. coli* J53 was used as a recipient bacterium, and *mcr-1*-harbouring strains were taken as a donor for mating-out assays on a filter. The experiment was performed by the method as defined in the published protocol. The transconjugants for *mcr-1* were selected on MacConkey agar plates supplemented with sodium azide (100 µg/ml) and colistin sulphate (2 µg/ml) (Ali et al., 2017). The transconjugants *E. coli* J53 were confirmed by antimicrobial susceptibility testing, *mcr-1* and *bla*<sub>CTX-M</sub> genes specific-PCR, and PCR-based replicon

typing to check its resistivity and the plasmid incompatibility (Inc) groups carried by the transconjugants.

## S1-pulsed-field gel electrophoresis (S1-PFGE) and southern hybridization

The six transconjugative *E. coli* J53 strains were selected and characterized by S1-PFGE and Southern hybridizations to determine the sizes of *mcr-1*-harbouring plasmids. The genomic DNA of each isolate was digested with S1 nuclease (Thermo scientific), and the PFGE was performed by following the previously described protocol (Shafiq et al., 2019). The southern blotting of plasmid DNA was carried out using the digoxin-labelled *mcr-1*-specific probe according to the manufacturer's instructions (Roche Diagnostics).

## RESULTS

### Bacterial isolation and culture results from food-producing animals

The 250 freshly passed faecal samples from different healthy food animals, including buffaloes ( $n = 50$ ), cattle ( $n = 50$ ), sheep ( $n = 50$ ), goat ( $n = 50$ ) and broilers ( $n = 50$ ), were collected from five local farms in Peshawar city to screen ESBL-positive and colistin-resistant *E. coli* isolates in livestock sector of Pakistan. Overall, 153 *E. coli* isolates were confirmed using conventional PCR, targeting the *uidA* gene from the collected samples. The positive rates of *E. coli* in buffaloes, cattle, sheep, goat and broilers were 60% (30 isolates/50 faecal samples), 74% (37 isolates/50 samples), 54% (27 isolates/50 samples), 50% (25 isolates/50 faecal samples) and 68% (34 isolates/50 samples), respectively. Among 153 *E. coli* isolates, 49.01% (75/153) and 18.95% (29/153) were ESBL producing and colistin resistant, respectively, as determined by phenotypic approaches. The prevalence of ESBL-producing and colistin-resistant *E. coli* obtained from each animal type is shown in Table 1.

**TABLE 1** Prevalence of ESBL-producing and colistin resistance among *Escherichia coli* isolates of different origins

| Animals   | No. of analysed samples | Number of positive samples (%) | Number of ESBL-Producing <i>E. coli</i> (%) | Number of colistin-Resistant <i>E. coli</i> (%) |
|-----------|-------------------------|--------------------------------|---------------------------------------------|-------------------------------------------------|
| Buffaloes | 50                      | 30 (60)                        | 14 (46.66)                                  | 02 (6.66)                                       |
| Cows      | 50                      | 37 (74)                        | 23 (62.16)                                  | 14 (37.83)                                      |
| Sheep     | 50                      | 27 (54)                        | 10 (37.03)                                  | 02 (7.40)                                       |
| Goat      | 50                      | 25 (50)                        | 08 (32.00)                                  | 0 (0)                                           |
| Broilers  | 50                      | 34 (68)                        | 20 (58.82)                                  | 11 (32.35)                                      |
| Total     | 250                     | 153 (61.2)                     | 75 (49.01)                                  | 29 (18.95)                                      |

## Antimicrobial susceptibility rate and patterns of *E. coli*

The antibiotic sensitivity profiles of 153 *E. coli* strains were obtained against 13 commonly used antibiotics, as shown in Table 2. The most common antimicrobial agents for which the strains showed a high resistant rate were tetracycline (100.0%), ampicillin (92.81%), trimethoprim (92.15%), gentamicin (67.97%), chloramphenicol (63.39%) and enrofloxacin (53.59%). The resistance rate to other antimicrobial agents, including ceftiofur (49.01%), followed by cefotaxime, was (47.71%), cefoxitin (45.09%), ciprofloxacin (43.13%), colistin (18.95%) and polymyxin-B (19.60%). All of the *E. coli* strains were sensitive to meropenem. Among all the isolates, 84.31% of the *E. coli* showed resistance to three or more antibiotics and were considered the MDR strains (Table 3). The most common pattern of MDR was observed for ampicillin, tetracycline and trimethoprim (84.31%).

## Molecular characterization of ESBLs and colistin-resistant genes

The 75 ESBL-positive *E. coli* strains were further selected for molecular characterization to screen the selective ESBLs-encoding and colistin resistance genes. The majority of ESBLs-encoding and *mcr-1* genes were found in the isolates from cattle and poultry sources. The positive rates of the ESBLs genes in buffaloes, cattle, sheep, goats and poultry were 46.66 (14/30), 62.16 (23/37), 37.03 (10/27),

32.00 (08/25) and 58.82% (20/34), respectively. Genotyping of these isolates via PCR determined that among ESBLs genes, *bla*<sub>CTX-M</sub> was the most prevalent group detected in 98.66% (*n* = 74/75) isolates, followed by *bla*<sub>TEM</sub> 49.33% (*n* = 37/75) and *bla*<sub>SHV</sub> 42.66% (*n* = 32/75). Further genotyping of *bla*<sub>CTX-M</sub> shows that *bla*<sub>CTX-M-1</sub> was the most prevalent group 47.29% (*n* = 35/74), followed by *bla*<sub>CTX-M-9</sub> 43.24% (*n* = 32/74), *bla*<sub>CTX-M-2</sub>, 6.75% (*n* = 5/74), *bla*<sub>CTX-M-25</sub> 4.05% (*n* = 3/74) and *bla*<sub>CTX-M-8</sub> 1.35% (*n* = 1/74). Meanwhile, in 29 colistin-resistant strains, only the *mcr-1* gene was identified in all of the isolates. The presence of *mcr-1* in the isolates of buffaloes, cattle, sheep and poultry were 6.66 (2/30), 37.83 (14/37), 7.40 (2/27) and 32.35% (11/34), respectively. No single colistin-resistant gene was detected in *E. coli* isolates collected from the goat. The distribution of ESBLs and *mcr-1* among different animals is presented in Figure 2.

## Co-occurrence of ESBLs and *mcr-1*-harbouring *E. coli*

Based on the results of this study, among 75 ESBLs-positive *E. coli* strains, 38.66% (29/75) were found to carry both *mcr-1* and ESBL genes. The *bla*<sub>CTX-M</sub> was the most prevalent gene, coexist with *mcr-1* at the rate of 38.66% (*n* = 29/75), followed by *bla*<sub>SHV</sub> and *mcr-1* 16% (*n* = 12/75), and *bla*<sub>TEM</sub> and *mcr-1* 14.66% (*n* = 11/75). The distribution of all *E. coli* strains co-harbouring ESBL and *mcr-1* genes isolated from different food-producing animals is presented in Figure S1(a-e).

**TABLE 2** Distribution of MICs of 13 antimicrobials for 153 *Escherichia coli* isolates of different origins

| Antibiotics     | MIC (mg/L) |      |     |    |    |    |    |    |    |    |     |     |      | Resistance |
|-----------------|------------|------|-----|----|----|----|----|----|----|----|-----|-----|------|------------|
|                 | <0.125     | 0.25 | 0.5 | 1  | 2  | 4  | 8  | 16 | 32 | 64 | 128 | 256 | >256 |            |
| Ampicillin      | –          | –    | –   | 1  | 1  | 2  | 2  | 5  | 12 | 19 | 31  | 41  | 39   | 92.81%     |
| Cefotaxime      | –          | –    | 32  | 25 | 23 | 16 | 13 | 7  | 5  | 8  | 13  | 4   | 7    | 47.71%     |
| Cefoxitin       | –          | –    | –   | 14 | 24 | 18 | 28 | 15 | 19 | 8  | 17  | 4   | 6    | 45.09%     |
| Ceftiofur       | 5          | 8    | 21  | 19 | 25 | 15 | 8  | 17 | 8  | 9  | 13  | 3   | 2    | 49.01%     |
| Chloramphenicol | –          | –    | –   | 1  | 1  | 2  | 13 | 39 | 25 | 19 | 22  | 17  | 14   | 63.39%     |
| Ciprofloxacin   | –          | 20   | 21  | 19 | 27 | 13 | 20 | 9  | 11 | 9  | 4   | –   | –    | 43.13%     |
| Colistin        | 27         | 29   | 41  | 27 | 5  | 8  | 9  | 3  | 2  | 2  | –   | –   | –    | 18.95%     |
| Enrofloxacin    | –          | 8    | 24  | 8  | 16 | 15 | 29 | 23 | 16 | 14 | –   | –   | –    | 53.59%     |
| Gentamicin      | –          | –    | 3   | 4  | 7  | 18 | 17 | 38 | 31 | 22 | 9   | 4   | –    | 67.97%     |
| Meropenem       | 106        | 22   | 11  | 14 | 0  | 0  | 0  | 0  | 0  | –  | –   | –   | –    | 00.00%     |
| Polymyxin-B     | 47         | 35   | 22  | 19 | 8  | 11 | 4  | 3  | 3  | 1  | 0   | –   | –    | 19.60%     |
| Tetracycline    | –          | –    | 0   | 0  | 0  | 0  | 0  | 31 | 23 | 26 | 28  | 21  | 24   | 100.0%     |
| Trimethoprim    | –          | –    | –   | –  | –  | 1  | 1  | 5  | 5  | 14 | 47  | 62  | 18   | 92.15%     |

Notes. Red vertical lines represent the breakpoints value between sensitive and resistant values. White areas show the drug dilutions used while the grey show that this range was not used for drug dilutions.

**TABLE 3** Multidrug-resistant patterns of *Escherichia coli* isolates recovered from food-producing animals

| No. of antibiotics | Resistance pattern                                              | No. of isolates | Resistance (%) |
|--------------------|-----------------------------------------------------------------|-----------------|----------------|
| 3                  | AMP, TET, TRI                                                   | 129             | 84.31          |
| 4                  | AMP, CHL, TET, TRI                                              | 110             | 71.89          |
| 5                  | AMP, CHL, GEN, TET, TRI                                         | 73              | 47.71          |
| 6                  | AMP, CEF, CHL, CTX, TET, TRI                                    | 49              | 32.02          |
| 7                  | AMP, CHL, ENR, CIP, GEN, TET, TRI                               | 32              | 20.91          |
| 8                  | AMP, CEF, CFX, CHL, CTX, CIP, TET, TRI                          | 29              | 18.95          |
| 9                  | AMP, CEF, CFX, CHL, CTX, COL, GEN, TET, TRI                     | 7               | 4.57           |
| 10                 | AMP, CEF, CFX, CHL, CIP, CTX, ENR, GEN, TET, TRI                | 3               | 1.96           |
| 11                 | AMP, CEF, CFX, CHL, CIP, COL, CTX, GEN, POL, TET, TRI           | 1               | 0.65           |
| 12                 | AMP, CEF, CFX, CHL, CIP, COL, CTX, ENR, GEN, POL, TET, TRI      | 1               | 0.65           |
| 13                 | AMP, CEF, CFX, CHL, CIP, COL, CTX, ENR, GEN, MEM, POL, TET, TRI | 0               | 0.00           |

Abbreviations: AMP, ampicillin; CTX, cefotaxime; CFX, ceftiofur; CEF, ceftiofur; CHL, chloramphenicol; CIP, ciprofloxacin; COL, colistin; ENR, enrofloxacin; GEN, gentamicin; MEM, meropenem; POL, polymyxin-B; TET, tetracycline; TRI, trimethoprim.

## Prevalence of mobile genetic elements and plasmid incompatibility groups

The prevalence of mobile genetic elements among *E. coli* strains, which play a key role in transmitting *mcr-1* and ESBL genes, was investigated. The observed insertion sequences were *ISECP1* 35.13% ( $n = 26/74$ ), followed by *ISCR1* 33.78% ( $n = 25/74$ ) and *ISApI1* 20.27% ( $n = 15/74$ ), while the *Int1* detected was 58.10% ( $n = 43/74$ ) among all ESBL-positive *E. coli* strains. Most of the  $\beta$ -lactamases encoding genes, such as *bla*<sub>CTX-M</sub>, *bla*<sub>TEM</sub> and *bla*<sub>SHV</sub>, were physically linked with these mobile elements. The insertion sequences found in *mcr-1*-positive *E. coli* strains were the *ISCR1* 34.48% ( $n = 10/29$ ), followed by *ISApI1* 31.03% ( $n = 9/29$ ) and *ISECP1* 24.13% ( $n = 7/29$ ). Mobile genetic elements identified in *E. coli* strains from different animals are presented in Figures 2 and 3.

PCR-based replicon typing (PBRT) confirmed that all ESBL and *mcr-1*-positive *E. coli* strains displayed plasmids of wide-range replicon types. The most prevalent replicon types were IncFIB 46.66% ( $n = 35/75$ ), followed by IncFIA 30.66% ( $n = 23/75$ ), IncFIC 26.66% ( $n = 20/75$ ), IncFrepB 26.66% ( $n = 20/75$ ), IncHI2 26.66% ( $n = 20/75$ ), IncP 22.66% ( $n = 17/75$ ), IncX4 21.33% ( $n = 16/75$ ), IncI1 18.66% ( $n = 14/75$ ), IncA/C 14.66% ( $n = 11/75$ ), IncN 8% ( $n = 6/75$ ) and IncY 2.66% ( $n = 2/75$ ) (Figure 3).

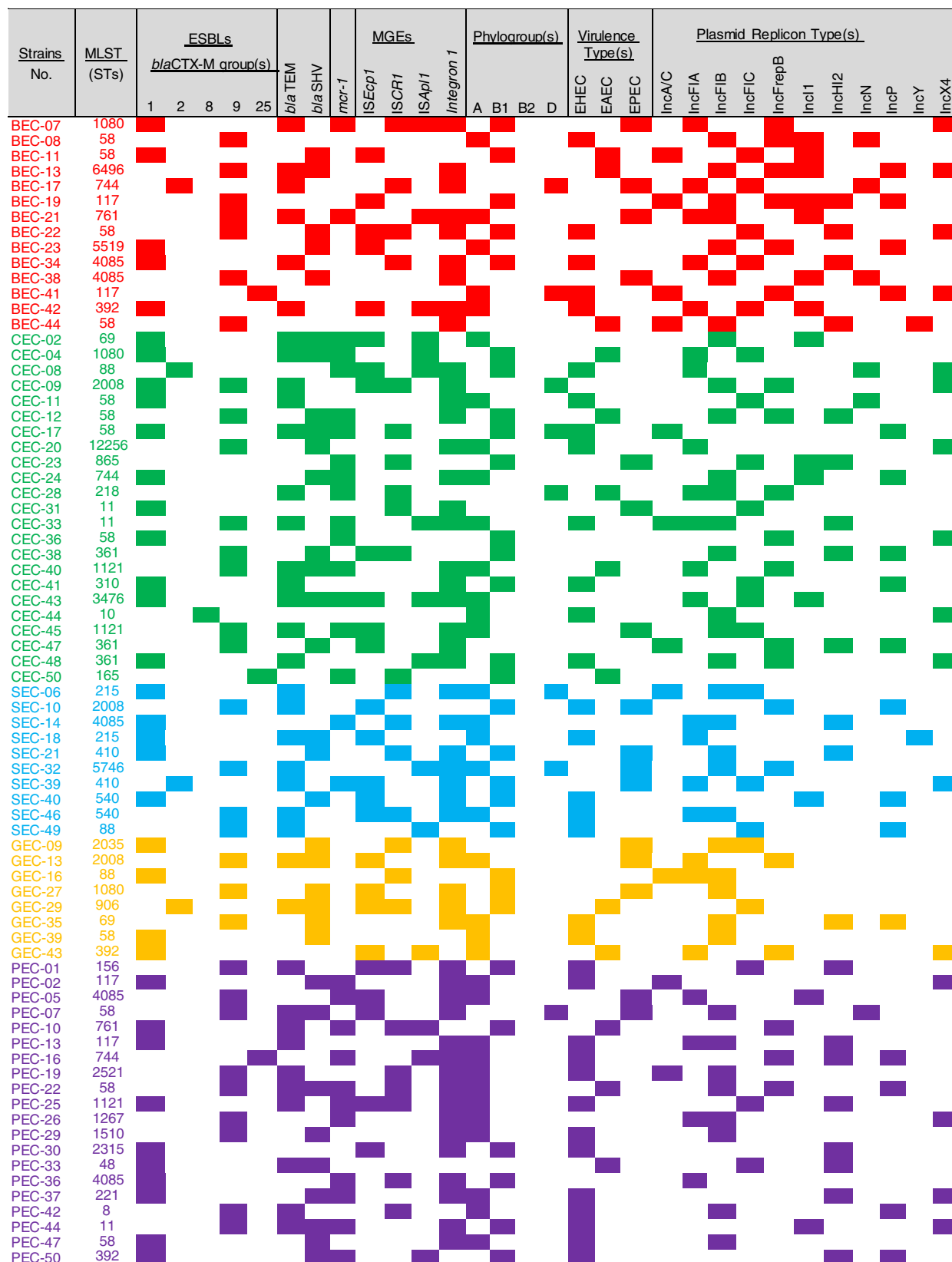
## Phylogenetic grouping and detection of virulence genes

The phylogenetic grouping of the 75 *E. coli* strains confirmed that phylogenetic group A was the dominant group 46.66% ( $n = 35/75$ ), followed by Phylogenetic group B1 38.66% ( $n = 29/75$ ), and another less common group D

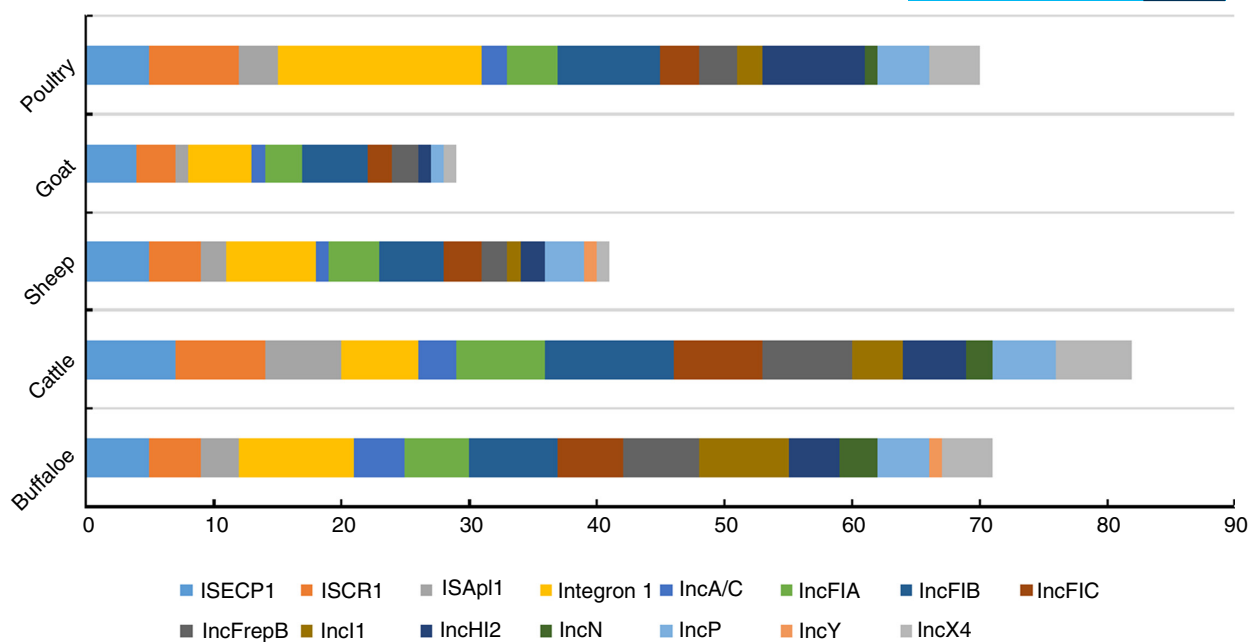
10.66% ( $n = 8/75$ ). None of the strains was found positive for phylogenetic group B2. All the isolated strains were further screened for the presence of various six diarrheagenic *E. coli* (DEC) virulence genes. The most prevalent DEC *E. coli* strains were EHEC 44% ( $n = 33/75$ ), followed by EPEC 21.33% ( $n = 16/75$ ) and EAEC 17.33% ( $n = 13/75$ ). Three other virulence gene types, DAEC, EIEC and ETEC, were not found. The findings and incidence of phylogenetic groups and virulence genes among *E. coli* strains from different animals are presented in Figure 2.

## Genetic characteristics and clonal relatedness of ESBL-harboursing *E. coli* among food animals

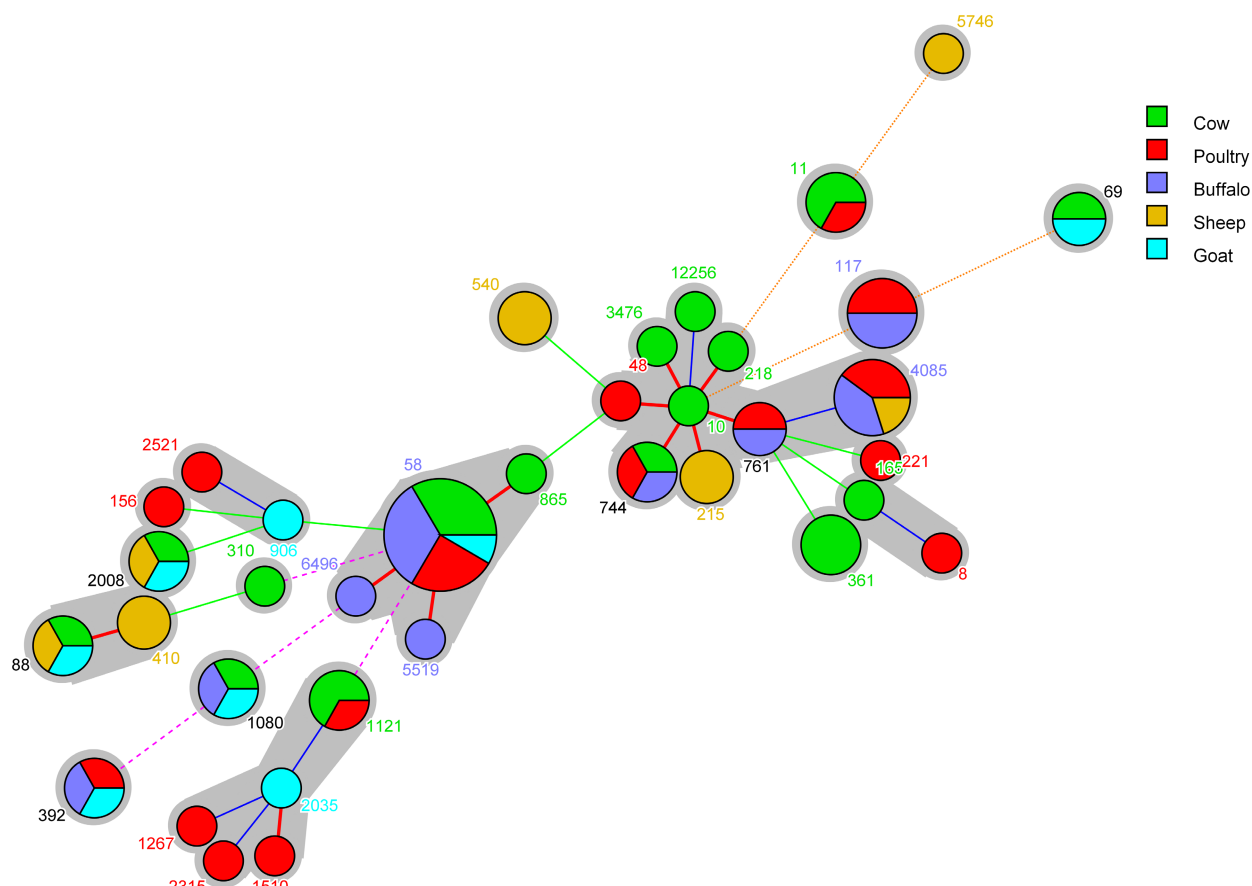
A total of 75 strains from different food-producing animals were organized into 56 sequence types (STs). Details of distribution and abundance of STs are labelled in Figure 4. ESBLs-positive *E. coli* strains from buffaloes belonged to nine different STs (four strains belonged to ST58, two strains were ST117 and ST4085, while one strain was ST1080 and ST6496 ST744, ST761, ST5519 and ST392). Two ESBL strains positive for *bla*<sub>CTX-M</sub> and *bla*<sub>TEM</sub> co-harboursing the *mcr-1* gene belonged to ST1080 and ST761, respectively. In all, 23 ESBLs *E. coli* strains from cattle belonged to 16 different STs (four were ST58, three were ST361, two were ST11, two others were ST1121, and one strain each was ST69, ST1080, ST88, ST2008, ST12256, ST865, ST744, ST218, ST310, ST3476, ST10 and ST65). Three ESBL *E. coli* strains from cattle co-harboursing *mcr-1* gene belonged to ST58. The *E. coli* strains isolated from sheep consist of seven different STs (two strains were ST215, two were ST410, two



**FIGURE 2** Distribution of sequence types (STs), antibiotic resistance genes (ARGs), mobile genetic elements (MGEs), phylogroups, virulence genes and plasmid replicons in 75 *Escherichia coli* strains recovered from food-producing animals. Colour boxes show the presence, and white boxes show the absence of the relevant gene in each *E. coli* strain. Strains indicated red were isolated from buffalo, green from cattle, blue represent strains from sheep, yellow from goats and isolates in purple were isolated from poultry. Blocks indicate the host association of the strain based on PCR



**FIGURE 3** Mobile genetic elements (MGEs) identified in *Escherichia coli* strains from food-producing animals



**FIGURE 4** Minimum spanning tree analysis of 75 *Escherichia coli* isolates based on allelic profiles of 7 housekeeping genes, using BioNumerics version (8.0). Each circle corresponds to a sequence type (ST), and the number of each ST is given beside the circle. The circle size depends on the number of isolates found within that profile, and the width of the line connecting two strains indicates the genetic variation between these isolates. The colour of the circles represents the source of the strains (green, cow; red, poultry; purple, buffaloes; yellow, goats; and blue, sheep). Grey shadows between some circles delineate that these strains belong to the same ST complex

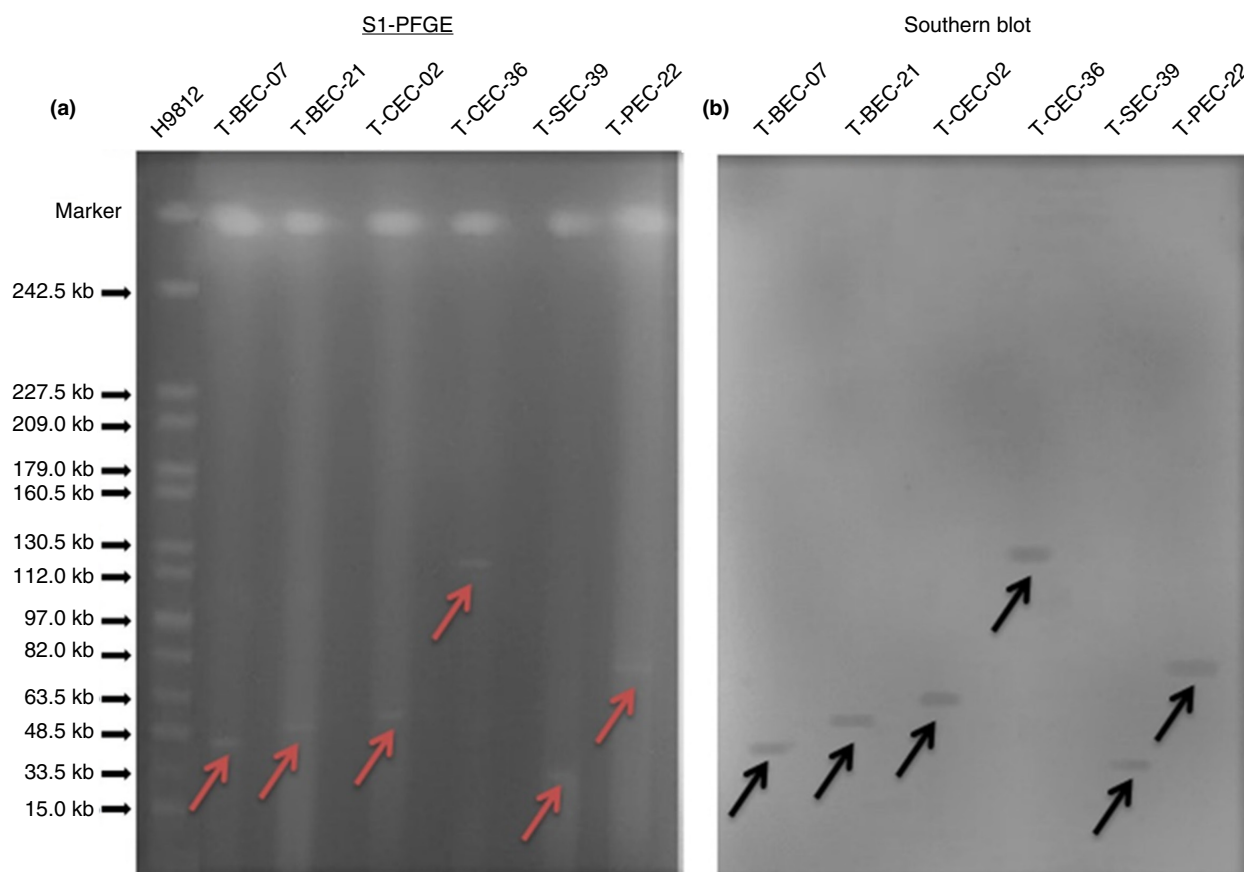
were ST540, while the remaining strains were ST2008, ST4085, ST5746 and ST88). In sheep, two ESBL-positive *E. coli* strains were found positive for *mcr-1* gene belonged to ST4085 and ST410. The STs of *E. coli* isolated from goats were different from each other (i.e. ST2035, ST2008, ST88, ST1080, ST906, ST69, ST58 and ST392). In all, 20 *E. coli* strains recovered from poultry belonged to 16 different STs (three were ST58, two of each strain were ST117 and ST4085, and one strain each was ST156, ST761, ST744, ST2521, ST1121, ST1267, ST1510, ST2315, ST48, ST221, ST8, ST11 and ST392).

Based on the MLST data of the chosen strains ( $n = 75$ ), the overall epidemiological evidence of the dominant STs indicated 56 STs. The most predominant sequence type found was ST58, 16% ( $n = 12/75$ ), followed by ST4085 6.66% ( $n = 5/75$ ) and ST117 5.33% ( $n = 4/75$ ). Among these, four ESBL strains harbouring *mcr-1* belonged to ST58, other three *mcr-1*-positive strains belonged to ST4085 and ST1121. While the remaining *mcr-1*-harbouring *E. coli* strains from different animals belonged to 18 different STs.

## Transconjugation and plasmid transferability analysis of *mcr-1* carrying strains

Conjugation experiments were performed on six randomly chosen *mcr-1*-positive *E. coli* strains to assess the transferability of the *mcr-1* gene among livestock. Out of the selected *E. coli* strains, all the six isolates were successfully transconjugated and transferred the *mcr-1* gene from donor strain to recipient *E. coli* J53. All the transconjugants harbouring the *mcr-1* gene were confirmed through antimicrobial susceptibility testing and PCR assays for the *mcr-1* gene. The MIC of colistin for all the transconjugants were twofold to fourfold increases compared to the donor strains. The remaining resistant profiles for all the other antibiotics, including the cephalosporins, were the same as that of the donor strains.

To determine the size, location and transferability of the *mcr-1* gene, S1-PFGE and southern blotting assays were carried out for six *E. coli* J53 transconjugants. S1-PFGE analysis indicated that the *mcr-1* gene located on



**FIGURE 5** (a) S1-nuclease pulsed-field gel electrophoresis profiles of six transconjugant *Escherichia coli* isolates and (b) southern blot hybridization of six *E. coli* transconjugants carrying the *mcr-1* on different plasmids with unlike positions with the size approximately 45 (IncFIC), 48 (IncFIB), 60 (IncFIA), 112 (IncA/C), 33 (IncX4) and 80 kb (IncFIB), respectively. Notes: Marker—*Salmonella* H9812 (15.0–242.5 kb). The red arrows show *mcr-1*-harbouring plasmids of transconjugants on the gel, while black arrows indicate *mcr-1*-bearing plasmids of the transconjugants on the nylon membrane

various plasmids, varying in sizes ranging from ~33-kb to ~112 kb [Figure 5a](#). Southern blotting further confirmed that the *mcr-1* gene was probed on different plasmids with unlike positions with the size approximately 45 (IncFIC), 48 (IncFIB), 60 (IncFIA), 112 (IncA/C), 33 (IncX4) and 80 kb (IncFIB), respectively, ([Figure 5b](#) in transconjugants). The plasmid types of the transconjugants were determined by the PBRT kit. In [Figure 5b](#), the red arrows show *mcr-1*-harbouring plasmids of transconjugants on the gel, while black arrows indicate *mcr-1*-bearing plasmids of the transconjugants on the nylon membrane.

## DISCUSSION

The misuse of antimicrobials not only in humans but, more importantly in food-producing animals is troublesome for public health. The increasing reports of MDR *Enterobacteriaceae* and ESBLs-and-colistin co-producers are highly alarming. Continuous national-wide surveillance studies and continuous monitoring are required to design an effective strategy to control the dissemination of MDR isolates. In Pakistan, the usage of antibiotics is not well-regulated resulting in overuse of antibiotics in society. Alongside this, no national wide consistent surveillance of antibiotic resistance is performed. In such a scenario, random studies are important for the scientific community and healthcare professionals. This study was designed to isolate *E. coli* from the faecal samples of healthy food animals to investigate ESBLs-encoding and colistin-resistant genes, along with the associated MGEs and virulence factors were performed.

This study reported 62% and 58% of ESBL-producing *E. coli* isolated from cattle and poultry, respectively, which is higher than the recent findings of ESBL-producing *Enterobacteriaceae* recovered from faecal specimens of animals in Pakistan (Ejaz et al., 2021; Umair et al., 2019) and lower than our previously reported ESBL-producing *E. coli* from bovine mastitis and chicken from China (Shafiq, Huang, Shah, Ali, et al., 2021; Shafiq, Huang, Shah, Wang, et al., 2021). The difference in isolation rate might be due to variation across populations in different regions, usage of antibiotics and size of the population. The high rate of ESBL-producing strains in healthy animals reported in the current study could be attributed to the extensive use of antimicrobials in farms, untrained workers, and keeping healthy and infected animals in close proximity (Ejaz et al., 2021).

We found that the isolates in our study were resistant to a range of antibiotics, including cephalosporins, aminoglycosides, fluoroquinolones, while intermediate resistance was observed against colistin and polymyxin-B. The MDR *E. coli* in the current study were 84%, which is

nearly similar to our previous findings from cattle (85%) and chicken (83%) samples (Shafiq, Huang, Shah, Ali, et al., 2021; Shafiq, Huang, Shah, Wang, et al., 2021). All isolates were resistant to tetracycline and ampicillin, while over 50% to trimethoprim and gentamicin. This high level of observed resistance pattern against a range of antibiotics may be attributed to a number of factors like selection pressure due to overuse of antibiotics and accumulation of MGEs in the environment as reported earlier (Brower et al., 2017; Ejaz et al., 2021; Younas et al., 2019).

We found that *bla*<sub>CTX-M</sub> was the most prevalent group at 98.66%, followed by *bla*<sub>TEM</sub> 49.33% and *bla*<sub>SHV</sub> at 42.66%. The prevalence of *bla*<sub>CTX-M</sub> and other ESBL genes in our study is relatively higher than in some earlier studies from Pakistan (Ejaz et al., 2021; Rahman et al., 2018; Umair et al., 2019), consistent with our previous reports from mainland China and other parts of the world (Shafiq, Huang, Shah, Ali, et al., 2021; Shafiq, Huang, Shah, Wang, et al., 2021). Further characterization of *bla*<sub>CTX-M</sub>-positive isolates revealed that *bla*<sub>CTX-M-1</sub> and *bla*<sub>CTX-M-9</sub> are the most prevalent genotypes. Previous studies also suggest that *bla*<sub>CTX-M-1</sub> is the most frequently occurring genotype recovered from multidrug-resistant and ESBL-producing isolates (Abbas et al., 2019; Shafiq, Huang, Shah, Wang, et al., 2021). It has been reported that *bla*<sub>CTX-M-1</sub> and *bla*<sub>CTX-M-9</sub> were the most common genotypes recovered from cattle in China (Ali et al., 2017; Shafiq, Huang, Shah, Wang, et al., 2021) which was similar to our study. Another study from poultry found a high incidence of *bla*<sub>CTX-M-9</sub> (89.3%), followed by *bla*<sub>CTX-M-1</sub> (10.7%), which was relatively higher in comparison with our present findings (Murtaza et al., 2021). Current findings may not be generalized due to a limited number of isolates from only one city of Pakistan, that is, Peshawar.

Our study indicates an overall 18.95% of colistin-resistant isolates recovered from healthy livestock comparatively lower than our previously reported study from China (Shafiq, Huang, Shah, Ali, et al., 2021; Shafiq, Huang, Shah, Wang, et al., 2021), whereas this incidence rate was higher than the recent reports from Pakistan (Ejaz et al., 2021; Lv et al., 2018). Furthermore, we report on *E. coli* colistin-resistant isolates for the first time from buffaloes (6.66%) and sheep (7.40%). Interestingly, all these resistant isolates were harbouring plasmid-mediated gene, *mcr-1* indicating its ability of fast dissemination and horizontal transfer. Previous reports from Pakistan have already detected the *mcr-1* gene from clinical, cattle, wild birds, poultry and the environment (Ali et al., 2021; Bilal et al., 2021; Mohsin et al., 2017, 2019). On top of this, the co-occurrence of ESBL and colistin is limiting the option for treatment in near future presenting a huge challenge for healthcare community.

Mobile genetic elements (MGEs) are vital in transmitting and circulating antibiotic resistance genes (ARGs) among bacteria. In this study, the most frequent genetic elements in ESBL-positive *E. coli* strains were *ISEcp1* 35.13%, followed by *ISCR1* 33.78%, and *ISapl1* 20.27%. Similarly, the *intI1* was detected 58.10% isolates, which has the ability to carry gene cassettes against multiple classes of antibiotics. The majority of the  $\beta$ -lactamases and *mcr-1*-positive *E. coli* were physically linked with these MGEs. Similar findings were ubiquitously reported in other studies from Pakistan and China indicated the significance of these MGEs in ARGs dissemination (Ali et al., 2020; Shafiq, Huang, Shah, Ali, et al., 2021). One of the interesting finding in our study was that the association of *mcr-1* with *ISCR1* 34.48% followed by *ISapl1* 31.03%. These results are contradictory with our previous report in chicken (Shafiq, Huang, Shah, Ali, et al., 2021), where *ISapl1* was the most prevalent IS element 51.85%, and considered a key element in propagating *mcr-1* gene. However, some other studies revealed that the *mcr-1* gene can be triggered by other MGEs or could emerge by itself independently (Petrillo et al., 2016; Zurfluh et al., 2016). Plasmid replicon typing analysis showed that all ESBLs and *mcr-1*-positive *E. coli* strains harbouring a wide range of plasmid types. The most predominant Inc. types found were IncFIB 46.66%, followed by IncFIA 30.66%, IncFIC 26.66%, IncFrepB 26.66%, IncHI2 26.66%, IncP 22.66% and IncX4 21.33%. Similar findings have been reported previously in which more than 10 different conjugative plasmids contributed to the spread of these resistant genes (Wang et al., 2017).

We were able to establish phylogenetic connections among isolates and within species of *E. coli* using MLST analysis. According to our findings, all 75 *E. coli* isolates were distributed into 56 sequence types (STs) with dominant STs observed were ST58, 16%, followed by ST4085 6.66% and ST117 5.33%. The most common genotypes of ESBL-producing *E. coli* in humans, such as ST131, were not found in this study. Among these, four ESBL strains harbouring *mcr-1* belonged to ST58, while other major positive *mcr-1* strains belong to ST4085 and ST1121. The remaining *E. coli* strains positive for *mcr-1* belonged to 18 different STs. The primary dominant ST58 identified here was similar to the previous studies in *mcr-1* *E. coli* from China and Brazil (Schaufli et al., 2016; Shafiq, Huang, Shah, Wang, et al., 2021). The remaining STs observed in our analysis were distinct from previously known STs of *mcr-1*-positive *E. coli* strains from humans and animals (Liu et al., 2020; Shen et al., 2018). As a result, we concluded that the present MLST data for *E. coli* were descriptive to the strain level, which cannot be used to determine the genetic background or origin of the isolates. We assumed that the strains were more likely linked

in evolutionary terms, raising concerns about the rapid spread between similar species.

Moreover, conjugation assays have proved that a number of different resistance genes and mobile elements could be transferred from donor to recipient bacteria. Genetic presentation of *mcr-1* carrying plasmids revealed that this gene could be found on different conjugative elements ranging in size from ~33 to ~112 kb. The conjugative characteristics of the plasmid backbone influence the transferability of *mcr-1* carrying plasmids between bacteria. The fact that *E. coli* isolates harbouring ESBL and *mcr-1* have unlike PFGE patterns implies that mobile elements play a critical role in transmitting *mcr-1*. Consequently, the current study shows that the persistent ESBL-producing and colistin-resistant *E. coli* carrying *mcr-1* in food-producing animals involves multiple genetic components, presenting a real threat to both animals and humans.

The limitation of the current study is that only a single isolate was recovered from each healthy animal, and the isolates were recovered from only one city in Pakistan. Thus, it is necessary to conduct other extensive country-wide studies on a wide range of food-producing animals better to analyse the prevalence and resistance mechanisms throughout Pakistan. However, this report is considered the first Pakistani study to assess the antimicrobial resistance in *E. coli* recovered from healthy food animals and found *mcr-1* in buffaloes and sheep sample sources for the first time.

In conclusion, we determined for the first time ESBLs and *mcr-1*-harbouring *E. coli* from the faecal samples of the healthy food-producing animals in Peshawar, Pakistan. Moreover, we found the diverse sequence and plasmid types, which showed that strong selection against the cephalosporins and colistin occurred in our isolates. The presence of ARGs in association with MGEs, as determined in the current study, is a severe threat to livestock and the human community as it has horizontally and food web transferability. Altogether, the current study indicates a high level of ESBL-producing *E. coli* co-harbouring *mcr-1* gene conferring resistance against the most valuable and last resort of antibiotics presenting a challenge to the healthcare community. The present study provided an alarming situation of drug usage in animal settings and strongly suggested that urgent interventions should be taken to minimize the rapid spread and emergence of colistin resistance in food animals. The high prevalence of ESBL and *mcr-1* in healthy food animals is an issue of serious concern given the opportunity of human transmission and the development of severe, untreatable infections. To counteract this problem, strict regulations such as prudent antimicrobial usage, large-scale monitoring and quick infection control strategies should be implemented.

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## CONFLICT OF INTEREST

The authors declare no conflict of interest.

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## SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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